

eRTMAC-NWIS

Nearby Wells Intelligence System

AI-Powered Offset Well Knowledge & Decision Support Platform for Drilling Operations

Combined Detailed Documentation & Project Report — SIH 2026 · Problem Statement ID 26121 · Oil India Limited

1. Executive Summary

eRTMAC-NWIS is a proposed AI-powered decision-support platform that gives drilling teams an institutional memory of historical offset-well experience. It connects what is happening in a current well with what happened previously in nearby or geologically similar wells, combining well-location data, drilling parameters, formation information, historical operational events, and unstructured reports such as Well Completion Reports (WCRs) and Daily Drilling Reports (DDR). AI-assisted document extraction, semantic search, GIS-based nearby-well analysis and a risk engine turn this scattered historical knowledge into searchable, correlated, actionable context.

The system complements — rather than replaces — Oil India Limited's existing eRTMAC real-time drilling monitoring environment. Oil India describes eRTMAC as a real-time monitoring and analytics environment that transmits critical well data to a command centre and supports real-time decisions. NWIS adds a historical and contextual intelligence layer: it asks what happened in comparable wells, where, at what depth/formation, how it was mitigated, and whether the current operation is approaching a previously problematic zone.

2. Problem Background & Core Problem

Modern drilling operations generate large volumes of time/depth-based measurements, logs, reports and operational knowledge. Oil India's published drilling and eRTMAC material shows the breadth of this information: real-time drilling data, geology and equipment data, drilling/well information repositories, comparative benchmarking, post-well reviews and drilling analytics are all relevant to decision support.

The deeper problem is not simply the absence of drilling data — it is that valuable drilling knowledge is distributed across structured databases, PDFs, scanned documents, logs and expert experience, and is difficult to retrieve, correlate and apply at the moment a decision is being made. A drilling engineer may know that a nearby well encountered mud loss, high torque, stuck pipe, pressure-related issues or cementing problems at a particular depth, but finding and correlating that information quickly is difficult without one searchable spatial system.

Core problem statement: Develop a unified Nearby Wells Intelligence System that maps nearby/historical wells, consolidates structured and unstructured well information, extracts key drilling events from reports, correlates current drilling conditions with historical wells by location/depth/formation, and provides contextual risk alerts and recommendations to support faster, evidence-based drilling decisions.

Typical fragmented workflow (today)

Active Well → Engineer → Search folders/PDFs → Ask colleagues → Find old report → Manually compare depths → Make decision

Proposed workflow (with NWIS)

Active Well → NWIS → Nearby/Similar Wells → Historical Evidence → Correlation → Risk Signal → Engineer Decision

Why the problem matters

- Historical drilling lessons can be reused instead of repeatedly rediscovered.
- Nearby wells provide valuable contextual evidence about formations and operational problems.
- Depth and formation correlation help engineers inspect relevant historical events before entering a known problem zone.
- Real-time monitoring is more useful when current measurements can be compared against historical context.
- A searchable repository reduces time spent manually opening and reading multiple reports.
- A spatial interface makes well-to-well comparison easier than document-only workflows.

3. Relationship with eRTMAC

eRTMAC	NWIS
Focuses on what is happening now	Focuses on what happened before and what historical context is relevant now
Real-time rig/well monitoring	Historical well intelligence + contextual correlation
Live drilling parameters	Historical parameters, events, formations and reports
Monitoring and analytics	Search, correlation, document intelligence and risk context
Decision support during active operations	Decision support enriched by nearby-well history

4. Core Idea

A drilling engineer is operating an active well and the bit is approaching a formation at a particular depth. The engineer opens the active well in NWIS. The system identifies nearby offset wells within a user-defined radius, and compares the active well's location, depth, formation and available drilling context with historical wells. It retrieves relevant incidents from nearby wells — mud losses, kicks, stuck pipe, torque spikes or cementing issues — and shows both the original evidence and the structured event information.

A similarity engine identifies wells with comparable geological/drilling conditions, and a risk engine generates a risk indicator when historical evidence suggests a potential problem. The alert explains the reason for the warning and provides relevant historical mitigation/lesson information, and the engineer can inspect the source reports before making an operational decision. The system therefore acts less like a generic chatbot and more like an intelligent operational memory for the drilling team.

5. Target Users

5.1 Drilling Engineers

- Open active-well context
- Find nearby wells
- Search historical events
- Compare wells by depth/formation
- Review drilling parameters
- Inspect source documents
- View risk indicators

- Receive proactive alerts
- Review historical mitigation measures

5.2 Drilling Supervisors / Field Personnel

- Monitor active operations
- View alerts
- Review relevant offset-well history
- Track recurring operational issues
- Access concise evidence before escalation

5.3 Office-Based Engineering Teams

- Search historical wells
- Analyze field/formation patterns
- Compare drilling performance
- Review lessons learned
- Generate operational intelligence

5.4 Administrators / Data Managers

- Manage users and roles
- Manage well metadata
- Monitor ingestion status
- Review extraction quality
- Manage document versions
- Monitor system/audit logs

5.5 Data/AI Teams

- Monitor extraction confidence
- Review model performance
- Manage embeddings/indexes
- Review risk-model versions
- Validate prediction outputs

6. Core Features

1. Nearby Well Intelligence — Interactive map showing nearby wells around the active well using a user-defined radius, with well metadata, trajectory and relevance indicators.

2. Historical Drilling Knowledge Repository — Central searchable repository of drilling events, lessons learned, operational challenges, mitigation actions and source documents.

3. AI Document Processing — OCR, document parsing, NLP extraction and normalization of information from historical drilling reports and PDFs.

4. Depth & Formation Correlation — Correlate current and historical wells using measured depth, formation, geological context and available drilling parameters.

- 5. Similar Well Discovery** — Rank historical offset wells by geographic, geological and operational similarity rather than distance alone.
- 6. Event Timeline** — Present incidents in chronological/depth order so engineers can understand what happened during previous drilling operations.
- 7. Risk Prediction** — Estimate potential mud losses, stuck pipe, overpressure, torque spikes and cementing issues using historical behaviour and available current parameters.
- 8. Proactive Alerts** — Generate alerts when the active well approaches a historically risky depth/formation or when current parameters resemble historical incidents.
- 9. AI Engineering Assistant** — Allow natural-language questions over structured events and source documents using grounded retrieval.
- 10. Evidence-Backed Recommendations** — Show relevant historical cases and mitigation information alongside each recommendation.
- 11. Well Comparison** — Compare selected wells side-by-side across trajectory, depth, formations, events and drilling parameters.
- 12. Operational Analytics** — Provide aggregate views of event frequency, depth ranges, formation-specific risks and recurring operational problems.
- 13. Realtime eRTMAC Adapter** — Provide an integration layer for current-well streams when an approved eRTMAC interface is available; otherwise support replay/simulation for the prototype.
- 14. Audit & Data Governance** — Track source documents, extraction versions, model versions, alert acknowledgements and important user actions.

7. Nearby Well & GIS / Spatial Intelligence

Map workflow: Active well selected → Radius selector → PostGIS spatial query → Nearby wells displayed → Engineer selects a well → Well intelligence panel opens.

- Display wells as points and, where available, directional trajectories as lines.
- Calculate wells within a selected radius such as 1 km, 5 km or 10 km.
- Filter nearby wells by field, formation, status, well type and drilling date.
- Highlight historical events along the current well's expected depth/formation path.
- Support depth-vs-distance and formation correlation views.

8. Historical Knowledge Repository & Document Intelligence Pipeline

The repository transforms scattered documents (digital PDFs, scanned PDFs, or mixed-format documents) into searchable operational knowledge. The processing pipeline is asynchronous and traceable:

Document upload/import → File validation and classification → OCR when required → Text/table extraction → Section and metadata detection → NLP entity/event extraction → Normalization to NWIS schema → Confidence scoring → Human review queue for low-confidence records → Knowledge repository + vector index.

Every extracted event preserves a link to its source document and page/section whenever possible.

Example extracted record: Well W1018, depth 3,180 m, Barail formation, partial mud loss, high severity, mitigation involving lost-circulation material (LCM). Exact values in the prototype are illustrative, not real operational measurements.

9. Depth & Formation Correlation, and Similar Well Discovery

The same event type can have very different significance at different depths or formations. NWIS correlates measured depth, true vertical depth (where available), formation, well trajectory, geological/reservoir context, drilling parameters, historical event type, severity, timing, and mitigation/outcome.

Example correlation

Well	Depth	Formation	Event
Well A	2,810 m	Formation X	Mud loss
Well B	2,845 m	Formation X	Severe mud loss
Well C	2,870 m	Formation X	No major loss
Active Well	2,850 m	Formation X	Approaching interval

Similar well discovery — relevance factors

Factor	Purpose
Geographic proximity	Identify local offset wells
Formation similarity	Compare the same geological interval
Depth overlap	Find comparable drilling intervals
Trajectory similarity	Account for well path
Drilling-parameter similarity	Compare operational behaviour
Historical-event similarity	Find wells with comparable incidents
Data completeness	Prefer wells with sufficient evidence

Distance alone is not sufficient — a well 2 km away may be less relevant than a well 8 km away drilled through the same formation with a similar trajectory and drilling context.

10. Event Timeline

Each well has a depth-aware operational timeline, e.g.:

- 2,300 m — Formation entered
- 2,510 m — Torque increase
- 2,670 m — Mud-loss warning
- 2,710 m — Lost-circulation event
- 2,715 m — Mitigation applied
- 2,760 m — Drilling resumed

The engineer can click an event and see structured fields plus the original source evidence.

11. AI / RAG Layer & Engineering Assistant

Capability	Implementation idea
Semantic report search	Embeddings + pgvector
Question answering	RAG over report chunks and structured event records
Event extraction	LLM/NLP structured JSON extraction
Report summarization	LLM with citations to source report/page
Similar-well retrieval	Spatial + formation + depth filters followed by semantic similarity
Recommendation support	Retrieve historical mitigation actions; show evidence rather than inventing procedures

Example assistant queries

- Show mud-loss incidents in nearby wells around this formation.
- What problems occurred between 2,700 and 2,900 m?
- Which nearby wells had stuck-pipe incidents?
- What mitigation was used for similar events?
- Which wells are most similar to the current well?
- Why has this risk alert been generated?

A critical design principle is traceability: the assistant retrieves evidence first and generates an answer from that evidence — it should not invent drilling history — and clearly distinguishes historical evidence from model-generated interpretation.

Evidence-backed recommendations — four layers

Layer	Example
Signal	Current well approaching historically risky interval
Evidence	Two similar wells experienced mud loss
Historical action	Mitigation recorded in the source report
Decision support	Review the historical case and applicable operating procedure

Final operational decisions remain with authorized drilling personnel.

12. Risk Engine

For the prototype, the system starts with transparent rules and similarity scoring; supervised ML is added only where enough historical labeled data exists.

Risk signals by type

Risk	Possible signals
Mud loss	Historical losses in same formation/depth, mud parameter changes, similar offset behaviour
Stuck pipe	Historical stuck-pipe intervals, torque/drag patterns, trajectory/formation similarity

Risk	Possible signals
Overpressure	Historical pressure indicators, kicks, drilling behaviour and formation context where available
Torque spike	Current torque trend versus historical comparable intervals
Cementing issue	Historical cementing problems, casing interval and formation context

Prototype rule-engine logic

Signal	Example prototype rule	Output
Historical depth overlap	Multiple nearby wells had an event within ± 50 m	Depth-zone warning
Formation match	Current and historical wells share formation	Higher relevance
Spatial proximity	Historical well is within selected radius	Higher relevance
Current parameter anomaly	Torque/pressure/mud-loss deviates from baseline	Live anomaly
Multiple supporting signals	Depth + formation + event + current anomaly	Elevated contextual risk

Example risk output (structured JSON)

```
{
  "risk_type": "mud_loss",
  "risk_level": "high",
  "depth_range": [2810, 2875],
  "confidence": 0.82,
  "evidence": [
    "well-A:event-17",
    "well-B:event-08"
  ],
  "recommended_review": "historical mitigation records"
}
```

Example alert record

Field	Value
Risk	Potential mud-loss zone
Current depth	2,850 m
Historical interval	2,810–2,875 m
Similar wells	Well A, Well B
Evidence	3 historical events
Severity	High
Model/rule	NWIS-Risk-v1
Action	Review historical mitigation and relevant drilling parameters

Alert lifecycle: Generated → Delivered → Viewed → Acknowledged → Resolved/Closed. An alert explains why it was generated rather than presenting an unexplained score.

Risk scores are decision-support indicators, not guaranteed predictions. Any production deployment would require domain validation, historical back-testing, uncertainty analysis and operational approval.

Optional future ML layer

Once sufficient labeled historical data is available, supervised or anomaly-detection models can estimate the probability of selected events. Candidate methods include XGBoost/Random Forest for tabular risk classification, Isolation Forest or autoencoders for anomaly detection, and temporal models for sequential drilling signals.

Such models should be trained and validated on representative organizational data rather than synthetic hackathon data.

13. Well Comparison & Operational Analytics

Comparison view supports 2–4 wells side-by-side:

Attribute	Active Well	Offset A	Offset B
Distance	—	3.2 km	5.8 km
Formation	X	X	X
Depth overlap	2,500–2,900 m	2,480–2,920 m	2,600–2,880 m
Mud-loss events	—	2	1
Stuck-pipe events	—	0	1
Torque spikes	Current	3	2

Operational analytics views

- Event frequency by formation
- Event frequency by depth range
- NPT distribution
- Most common mitigation actions
- Field-level risk hotspots
- Well-to-well drilling parameter comparisons
- Document coverage and extraction quality
- Alert frequency and acknowledgement time

14. Real-Time Integration & Simulation

The PS positions NWIS alongside eRTMAC and lists eRTMAC data streams among potential OIL data sources. The prototype exposes a clean adapter interface; if a live interface is not supplied, a replay service emits timestamped active-well parameters through the same interface.

Proposed flow: eRTMAC / Replay Feed → Adapter → Validation → Active-Well State → Risk Engine → Alert Engine → Dashboard.

Actual eRTMAC integration may not be available to a student team. The prototype can simulate a live drilling stream with generated timestamped records — updating depth, torque, pressure and mud-loss values every few seconds — then trigger historical-context checks as the simulated well enters a risk zone.

15. Audit & Data Governance

- Source-document ID retained for extracted facts
- Document versioning
- Extraction model/version stored
- Risk model/version stored

- Alert acknowledgement audit
- User action audit
- Role-based access
- No silent overwrite of historical evidence
- Original source preserved

16. System Architecture

Overall dashboard/service architecture

FIELD / OFFICE USER → React + Tailwind + GIS Dashboard → REST API / WebSocket Layer → Node.js + Express Backend → PostgreSQL + PostGIS & AI/Intelligence Services (well/trajectory/event store, OCR + NLP, formation/parameters, embeddings/vector search, alerts/users/audit, RAG/similarity, risk analytics) → eRTMAC / Replay Adapter.

Object/document storage sits alongside the structured database for original reports and processed artifacts.

AI service data flow

React → Node/Express → AI Service → OCR / NLP / Embeddings / RAG / ML → Structured JSON + evidence references → Backend validation → Frontend.

The backend validates schema, permissions and evidence references before displaying any AI-generated result.

High-level pipeline architecture (alternative framing)

WCR/DDR/PDF + well datasets + simulated/current drilling stream → OCR / PDF extraction / validation → Structured event extraction + document chunks → PostgreSQL + PostGIS + pgvector → Spatial correlation + RAG + risk engine → FastAPI services / WebSockets → React + MapLibre dashboard.

Core dashboard screens

Screen	Main contents
Overview	Active wells, current depth, alerts, recent events
Well Map	All wells, nearby search, filters, trajectories
Well Detail	Depth, formation, drilling charts, event timeline
Historical Correlation	Nearby wells, matching depth/formation, event comparison
Document Intelligence	Upload/search reports, extracted entities/events
AI Well Assistant	Natural-language questions with source citations
Risk Monitor	Current parameters, risk zones, evidence and alerts

17. Recommended Technology Stack

The two source documents propose slightly different stacks for the backend, GIS library and AI service layer. Both are presented below — either is workable; the FastAPI variant keeps AI/ML code and API in one Python codebase, while the Node/Express variant keeps a single JavaScript codebase across frontend and backend.

Stack A — Node/Express-centric (PS-aligned)

Layer	Recommended technologies
Frontend	React, Tailwind CSS, React Router, Axios
GIS	Mapbox GL JS or Leaflet
Backend	Node.js, Express.js, REST APIs
Authentication	JWT + role-based authorization
Database	PostgreSQL + PostGIS
Vector search	pgvector or approved vector database
AI service	Python + FastAPI (sidecar service)
OCR	Tesseract/PaddleOCR or approved OCR service
NLP/LLM	Transformer/LLM API + structured extraction
ML	Python, scikit-learn/PyTorch as appropriate
Realtime	WebSocket or Server-Sent Events
Storage	S3-compatible object storage or approved equivalent
Deployment	Docker-based services where permitted

Stack B — Python/FastAPI-centric

Layer	Technology	Reason
Frontend	React + TypeScript	Fast, component-based dashboard development
UI	Tailwind CSS + shadcn/ui	Rapid polished UI
GIS	MapLibre GL JS	Interactive web mapping
Charts	Plotly	Technical drilling charts
Backend	Python + FastAPI	Fast API development + AI ecosystem
Database	PostgreSQL + PostGIS	Relational + spatial data
Vector search	pgvector	Keep semantic search inside PostgreSQL
PDF	PyMuPDF	Fast PDF text extraction
OCR	PaddleOCR	Scanned document extraction
RAG	LlamaIndex	Document indexing/retrieval workflow
AI/NLP	LLM + Hugging Face/spaCy as needed	Extraction, summarization, semantic analysis
ML	Scikit-learn + XGBoost	Prototype risk/anomaly models
Geo processing	GeoPandas + Shapely	Spatial calculations
Realtime	WebSockets	Live dashboard updates
Cache/jobs	Redis + Celery (optional)	Background processing

Layer	Technology	Reason
Deployment	Docker	Reproducible demo deployment

18. Database Design

Core schema (important fields)

Table	Important fields
users	id, name, email, password_hash, role, status, created_at
wells	id, well_name, field, latitude, longitude, status, spud_date, total_depth
well_trajectories	id, well_id, measured_depth, tvd, latitude, longitude, inclination, azimuth
formations	id, name, description, geological_attributes
well_formations	id, well_id, formation_id, top_depth, bottom_depth
drilling_parameters	id, well_id, timestamp, depth, wob, rpm, torque, rop, mud/pressure fields
drilling_events	id, well_id, depth, formation_id, event_type, severity, description, start_time, end_time
mitigations	id, event_id, action, outcome, notes
documents	id, well_id, document_type, file_uri, document_date, version, ocr_status
document_chunks	id, document_id, page, section, text, embedding, confidence
extracted_entities	id, document_id, entity_type, value, confidence, source_location
risk_predictions	id, well_id, depth, risk_type, score, evidence_refs, model_version, created_at
alerts	id, well_id, risk_type, severity, message, evidence_refs, status, acknowledged_by, timestamps
audit_logs	id, user_id, action, resource_type, resource_id, timestamp

System inputs / dataset requirements (production)

Dataset	Important fields	Use
Well master	Well ID, lat/long, field/block, type, status, dates	Map and well identity
Directional survey	MD, inclination, azimuth, TVD	Well trajectory / spatial comparison
Drilling time-series	Depth/time, ROP, WOB, RPM, torque, pressure, flow, mud parameters	Current-state analytics and anomaly detection
Formation/geology	Formation, top/bottom depth, lithology, markers	Depth/formation correlation
Historical events	Well, depth, event type, severity, description, mitigation	Risk rules and historical comparison
WCR/DDR/PDFs	Reports, logs, narrative observations	AI extraction and RAG

Dataset	Important fields	Use
Well logs	Depth-indexed log curves / interpreted properties	Advanced correlation
Real-time feed	Timestamped current well measurements	eRTMAC-style live dashboard

Prototype dataset (hackathon scale)

For a two-day prototype, a small synthetic dataset is sufficient — clearly labeled as simulated/prototype data. A practical starter package: 30–100 wells, 10–50 thousand drilling-parameter rows, 50–200 historical events, several formation records, and 10–20 sample WCR/DDR-like PDFs.

File	Suggested fields
wells.csv	well_id, latitude, longitude, field, status, well_type, total_depth
drilling_data.csv	well_id, timestamp, depth, ROP, WOB, RPM, torque, pressure, mud_flow, mud_loss
events.csv	event_id, well_id, depth, formation, event_type, severity, description, mitigation
formations.csv	formation, top_depth, bottom_depth, lithology, risk_factors
reports/	sample WCR/DDR PDFs for OCR/RAG demonstration

19. API Structure

Endpoint groups: /api/auth, /api/users, /api/wells, /api/wells/nearby, /api/wells/:id/trajectory, /api/wells/:id/formations, /api/wells/:id/events, /api/wells/:id/parameters, /api/documents, /api/documents/:id, /api/search, /api/assistant, /api/compare, /api/risks, /api/alerts, /api/realtime, /api/dashboard, /api/audit.

Example endpoints

```
GET /api/wells/nearby?lat=...&lon=...&radius=10
GET /api/wells/:id/events?from_depth=2500&to_depth=3000
POST /api/search
POST /api/assistant/query
POST /api/compare
GET /api/risks/:wellId
GET /api/alerts/:wellId
PATCH /api/alerts/:id/status
POST /api/documents/ingest
```

20. Authentication & Roles

A user should only access wells, documents and operational information authorized by the deployment's access policy.

Role	Main permissions
Drilling Engineer	Active-well view, nearby wells, search, comparison, risks, alerts, evidence
Supervisor	Engineer functions + broader operational monitoring and alert oversight

Role	Main permissions
Data Administrator	Well/document metadata, ingestion status, data-quality workflows
AI/ML Administrator	Model/index management, evaluation and extraction review
System Administrator	Users, roles, configuration, audit and system health

21. AI Safety, Reliability & Key Risks

AI should

- Retrieve historical evidence
- Summarize source documents
- Identify similar wells
- Explain risk signals
- Clearly communicate uncertainty
- Expose source references
- Support engineers with decision context

AI should not

- Invent historical events
- Claim unsupported operational certainty
- Hide the evidence behind a score
- Replace authorized engineering judgment
- Silently alter source records
- Present synthetic/demo data as real OIL data

Key project risks and mitigations

Risk	Mitigation
Poor/limited historical data	Start with a controlled prototype dataset; design ingestion for real data later
OCR errors	Show extracted text, confidence/validation steps, and allow correction
Hallucinated AI answers	RAG-only answering, source citations and structured data grounding
False alarms	Transparent rules, thresholds, evidence display and human review
Synthetic data mistaken for real data	Clearly label demo/simulated values
Sensitive operational data	Authentication, authorization, encryption and audit logging in production

Important operational principles

- NWIS is a decision-support system, not an autonomous drilling controller.
- Historical evidence and AI predictions must remain distinguishable.
- Recommendations should be reviewed by authorized drilling personnel.
- Source documents should remain accessible.

- Model outputs should include confidence and evidence coverage where possible.
- Data-quality limitations should be visible rather than hidden.

22. Data Privacy & Security

- Use authentication and role-based authorization.
- Use HTTPS/TLS in deployment.
- Hash passwords using an approved password-hashing algorithm.
- Validate uploaded documents and payloads.
- Restrict access to proprietary drilling reports.
- Maintain audit logs for important actions.
- Store secrets only in environment/secret-management systems.
- Separate demo/synthetic data from production data.
- Do not expose raw operational data through public endpoints.
- Track document and model versions.

This project describes a prototype architecture and should not be treated as a drilling procedure or operational safety system. Actual drilling decisions must remain under qualified personnel and applicable company procedures. For a hackathon demonstration, synthetic data and simulated live streams are appropriate; real confidential operational data should only be used with authorization.

23. Team Division & Deliverables

Member	Primary role	Responsibilities
Member 1	Backend + Database Lead	Node/Express, PostgreSQL/PostGIS, schema, migrations, auth, RBAC, well/event APIs, nearby-well queries, core integration
Member 2	Frontend + GIS Lead	React, Tailwind, GIS map, active well, nearby wells, trajectories, filters, well intelligence, comparison UI
Member 3	Realtime + Alerts + DevOps	eRTMAC/replay adapter, realtime state, WebSocket/SSE, alert engine, deployment, logging, environment/configuration
Member 4	AI/ML — Document Intelligence	OCR, PDF parsing, section detection, entity/event extraction, normalization, confidence scoring, document indexing
Member 5	AI/ML — RAG + Risk	Embeddings, semantic retrieval, reranking, RAG assistant, similar-well engine, risk baseline/model evaluation
Member 6	Analytics + QA + Integration	Cross-well analytics, risk dashboards, integration testing, data-quality tools, end-to-end testing, demo dataset, documentation/presentation

Deliverable ownership

Deliverable	Owner	Support
Database + PostGIS	M1	M6
Authentication/RBAC	M1	M3

Deliverable	Owner	Support
Well APIs	M1	M3
Nearby-well geospatial query	M1	M2
GIS dashboard	M2	M6
Well comparison UI	M2	M6
Realtime adapter	M3	M1/M5
Alert service	M3	M5/M6
Document ingestion	M4	M3
OCR pipeline	M4	M6
NLP extraction	M4	M5
Vector index	M5	M4
RAG assistant	M5	M4
Similar-well ranking	M5	M6
Risk engine	M5	M6
Analytics	M6	M1/M5
Testing	M6	All
Deployment	M3	M1
Final demo	M6	All

24. GitHub Workflow

Recommended branches: main, develop, feature/backend-schema, feature/gis-dashboard, feature/realtime-alerts, feature/document-ai, feature/rag, feature/risk-engine, feature/analytics.

Each developer works through a feature branch and Pull Request. Stable code is merged into develop; main contains release-ready versions.

Example workflow: git checkout -b feature/document-ai → implement → git add . → git commit -m "Add drilling event extraction" → git push → Pull Request → develop.

25. Development Plan

Phased plan

Phase	Main work
Phase 1 — Foundation	GitHub repository, architecture, database, authentication, React shell, sample well dataset, API contracts
Phase 2 — Core Well Intelligence	Well CRUD, PostGIS nearby search, trajectory visualization, formation/event schema, well detail page

Phase	Main work
Phase 3 — Document Intelligence	PDF ingestion, OCR, parsing, structured extraction, source references, document search
Phase 4 — RAG & Similarity	Embeddings, vector search, hybrid retrieval, similar-well ranking, AI assistant
Phase 5 — Risk & Alerts	Rules/similarity baseline, risk APIs, alert engine, evidence cards, acknowledgement lifecycle
Phase 6 — Realtime & Analytics	Replay/eRTMAC adapter, active-well updates, trends, cross-well analytics, risk dashboard
Phase 7 — Integration & QA	End-to-end testing, security, performance, extraction evaluation, hallucination checks, UI polish
Phase 8 — Presentation	Freeze build, prepare demo scenario, architecture diagram, pitch, screenshots, video and documentation

Condensed two-day hackathon schedule

Time	Deliverables
Day 1 — Morning	Project setup, React dashboard, database schema, synthetic dataset
Day 1 — Afternoon	Map, well markers, nearby-well search, well detail page
Day 1 — Evening	Events, formations, drilling charts, simulated live stream
Day 2 — Morning	PDF extraction, sample reports, vector indexing, RAG assistant
Day 2 — Afternoon	Risk rules, depth/formation correlation, alerts
Day 2 — Evening	UI polish, demo flow, testing, PPT screenshots and backup demo

26. MVP Priority

P0 — Must have

- Authentication/RBAC
- Well database
- Nearby-well GIS map
- Well detail page
- Historical event repository
- PDF/document ingestion
- Basic OCR/extraction
- Search by well/depth/formation/event
- Evidence-linked RAG assistant
- Similar-well ranking
- Basic risk indicators
- Alert dashboard

P1 — Should have

- Realtime replay
- Cross-well parameter charts
- Advanced risk model
- Advanced document extraction
- Well comparison
- Alert acknowledgement
- Operational analytics
- Admin/data-quality dashboard

P2 — Nice to have

- Actual eRTMAC integration if interface is provided
- Advanced deep-learning models
- 3D trajectory visualization
- Advanced predictive analytics
- Voice assistant
- Automated report generation
- Enterprise SSO integration

What NOT to build in two days

- A full production eRTMAC replacement.
- A physically validated drilling simulator.
- A large-scale deep-learning model trained from scratch.
- Real-time integration with confidential OIL systems without authorization.
- A complex microservice/Kafka/Kubernetes architecture unless the team already has it ready.
- Automatic operational recommendations presented as authoritative drilling instructions.

27. Definition of Done

Nearby Well Intelligence

- Engineer can select active well
- Radius query works
- Nearby wells appear on map
- Well metadata opens
- Trajectory can be displayed

Document Intelligence

- Document can be ingested
- OCR/parsing works
- Events/entities are extracted
- Source location is preserved
- Low-confidence results are flagged

RAG

- Question accepted
- Relevant evidence retrieved
- Answer follows schema
- Source references displayed
- Unsupported answers are avoided

Risk Engine

- Input accepted
- Risk calculation executes
- Evidence is attached
- Model/rule version stored
- Alert can be generated and acknowledged

Realtime

- Replay/live adapter emits data
- Active state updates
- Risk engine receives updates
- Alert reaches dashboard

Frontend

- All core API states handled
- Loading/error states work
- Map interactions work
- Evidence can be opened
- Role restrictions enforced

28. Main Demo Scenario

The final presentation should revolve around ONE active drilling well:

- Engineer logs into NWIS.
- Engineer opens the active well and sees current depth/formation.
- GIS map displays nearby offset wells inside a selected radius.
- Engineer selects the most relevant historical offset well.
- The system displays its trajectory, formations and event timeline.
- Engineer filters for mud-loss/stuck-pipe events in the relevant depth interval.
- NWIS retrieves source evidence from historical reports.
- Engineer asks the AI assistant what similar problems occurred.
- The RAG assistant returns a grounded answer with source references.
- The active-well replay advances toward a historically risky interval.
- The risk engine generates an alert.

- The alert shows the historical wells/events supporting it.
- Engineer opens the comparison screen.
- The system shows historical mitigation/lesson information.
- The dashboard summarizes the active risk and supporting evidence.

Example demo data

Field	Example / Demo value
Active Well	WELL-A-102 / NWIS-DEMO-01
Latitude/Longitude	Configured from well metadata
Current Depth	2,850 m
Current Formation	Example Formation X
Search Radius	10 km
Nearby Wells	12
Similar wells	3
Historical mud-loss cases	2
Historical stuck-pipe cases	1
Current risk	High — example/simulated
Evidence	3 linked historical events

All hackathon demo data should be synthetic or authorized. Do not present invented data as actual OIL operational records.

29. What Makes the Project Different

- Institutional memory: historical drilling experience becomes searchable rather than remaining locked in documents.
- Geospatial context: nearby wells are automatically connected to the active well.
- Depth-aware intelligence: events are correlated to the actual drilling interval.
- Formation-aware intelligence: historical behaviour is interpreted in geological context.
- Evidence-first AI: answers and alerts point back to historical evidence.
- Proactive decision support: the system surfaces relevant history before or while a risk develops.
- Operational workflow: the product is designed around an engineer's active-well decision, not a generic document chatbot.

Demo moments that land well

- Click a well on the map and immediately see its historical intelligence.
- Show a current depth marker moving through formations.
- Automatically surface nearby wells that experienced similar events.
- Ask a natural-language question and receive an evidence-linked answer.
- Trigger a contextual warning when the simulated current well approaches a historical event zone.

- Open the underlying report/event record to demonstrate traceability.

30. Production-Scale Evolution

- Connect authorized eRTMAC streams through secure interfaces.
- Integrate WITSML/industry-standard well-data interfaces where applicable.
- Add enterprise identity, RBAC, audit logs and encryption.
- Introduce data-quality monitoring and lineage.
- Expand from rule-based risk to validated ML models.
- Add well-log and directional-survey correlation.
- Add expert feedback loops so engineers can confirm or reject alerts.
- Back-test alerts against historical wells and measure false positives/false negatives.
- Deploy on approved on-premise/private infrastructure where data governance requires it.

31. Success Metrics & Expected Benefits

Prototype success metrics

Metric	Prototype target
Nearby-well search	Results returned in under a few seconds
Report extraction	Correctly extract key fields from sample documents
RAG answer grounding	Answers linked to retrieved source records
Risk explanation	Every alert shows why it triggered
Map interaction	Well → detail → historical event in a few clicks
Live simulation	Dashboard updates continuously without manual refresh

Expected benefits

Expected value	How NWIS supports it
Faster decision support	Relevant historical wells/events surfaced automatically
Reduced manual searching	OCR/NLP and searchable knowledge repository replace manual PDF hunting
Better use of institutional knowledge	Lessons from older wells become reusable
Proactive risk awareness	Depth/formation and similarity-based alerts surface potential problems
Better operational context	GIS + well trajectory + event timeline provide one view
Evidence-based decisions	Alerts and AI answers link back to source records
Scalable intelligence	The same architecture can process additional wells/documents over time

32. One-Line Project Pitch

eRTMAC-NWIS is an AI-powered institutional-memory and decision-support platform that turns historical offset-well reports and drilling data into searchable, geospatially correlated and proactive risk intelligence for drilling engineers.

33. Main Goal

The success of NWIS should not be measured by the number of AI features alone. It should be measured by whether an engineer can open an active well, immediately understand which nearby historical wells matter, retrieve the relevant drilling experiences, see comparable depth/formation behaviour, receive an evidence-backed risk signal, and inspect the source information before making an operational decision.

34. Source Boundary, Assumptions & Reference Basis

This documentation uses SIH Problem Statement 26121 as the source for the problem scope, requested capabilities and potential OIL data sources. The problem statement identifies WCRs, DDRs, drilling/mud-logging databases, historical well parameters, reservoir/geological data, eRTMAC streams, well trajectory/survey data, casing/cementing/mud records, and historical operational events as potential sources. It does not specify the exact production APIs, data schemas, dataset sizes, access permissions, accuracy targets, or a mandated technology stack — those implementation details are proposed architecture and should be validated against organizer-provided interfaces and data.

Public Oil India material supports this report's description of eRTMAC and the types of drilling information involved. Oil India states that eRTMAC supports real-time transmission and analysis of critical well data, and describes access to drilling, geology and equipment data, repositories of drilling/well information, comparative benchmarking, post-well reviews and drilling analytics. Oil India tender/report documents also show examples of well completion/final-well reporting, depth databases, gas data, lithology, cutting descriptions, mud logs, drilling logs, pore-pressure logs, deviation records and daily drilling information.

Sources used for background verification

- <https://www.oil-india.com/drilling>
- <https://www.oil-india.com/leveraging-technology>
- https://www.oil-india.com/files/oldtender/national/NIC_CDH6605P21.pdf
- https://www.oil-india.com/files/oldtender/national/NIT_CBI0213P23.pdf

35. Problem Statement Reference Details

Field	Detail
Organization	Oil India Limited
Department	Oil India Limited
Category	Software
Theme	Smart Automation
Primary users	Drilling engineers, field personnel, office-based drilling teams, supervisors
System role	Standalone decision-support platform alongside eRTMAC

Knowledge types captured

Knowledge type	Examples
Drilling events	Mud loss, kick, stuck pipe, fishing, NPT
Parameters	Torque, WOB, RPM, ROP, mud parameters, pressure fields where available
Formation context	Formation name, depth interval, geological attributes
Casing/Cementing	Casing program, cementing practices, problems and outcomes
Mitigation	Actions taken, materials/operations used, result
Lessons learned	Historical observations and recommendations
Source evidence	Document, page, section and extraction metadata

36. Final Implementation Checklist

- Architecture frozen
- Database schema migrated
- RBAC working
- Well map working
- Nearby-well query working
- Well/event data populated
- Document ingestion working
- OCR tested
- NLP extraction tested
- Source references preserved
- Vector search working
- RAG assistant grounded
- Similar-well ranking working
- Risk baseline working
- Alerts working
- Realtime replay working
- Analytics working
- Security checks complete
- End-to-end demo tested
- Presentation and technical documentation complete